AI-POWERED CROP DISEASE DETECTION SYSTEM

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A Project Proposal

PRESENTATION OUTLINE

- Abstract
- Introduction & Problem Statement
- Objectives
- Methodology (Agile + ML Pipeline)
- System Architecture (Web + API)
- Use case diagram
- DFD (0, 1)
- Database Design (ER Diagram)
- Cost Estimation (COCOMO II)
- Project Schedule
- Project Deliverables
- Conclusion
- References & Q&A

ABSTRACT

- Crop diseases cause 20-40% yield losses (>$220 billion)
- Delayed diagnosis in rural areas is the main problem
- **Solution:** AI-based – upload leaf image → instant disease ID + treatment
- MobileNetV2 trained on PlantVillage (38 classes, 14 crops, 90-95% accuracy)
- Web + REST API (Flask, JWT, MySQL)
- Cost: ≈ ₹4.07 lakhs (COCOMO II)

INTRODUCTION

- Agriculture feeds most of the world, but diseases destroy 20-40% of crops
- Farmers cannot identify diseases early or correctly – no local experts
- **Our system:** Farmer takes leaf photo → uploads via web browser → instant diagnosis + treatment
- Three parts:
    - Deep learning model (38 classes)
    - Web frontend (HTML/CSS/JS)
    - Flask backend + MySQL + REST API
- Features: scan history, OTP password reset, admin panel

PROBLEM STATEMENT

- No rapid, low-cost tool for accurate disease identification
- Manual diagnosis = slow and error-prone
- No integrated platform for detection, history, treatment lookup
- Existing web tools lack proper user management & admin dashboard

OBJECTIVES

A) General Objective

Develop an AI-powered crop disease detection system (web + REST API) with treatment recommendations.

B) Specific Objectives

- MobileNetV2 model (38 diseases, $\geq$85% accuracy)
- Flask REST API (JWT auth, prediction, history, admin)
- Responsive web frontend + admin dashboard
- Disease-specific treatments (organic/chemical), prevention, severity
- Scan history, OTP password recovery, admin user management

LITERATURE REVIEW – KEY STUDIES

	Approach		
Mohanty et al. (2016)	CNN on PlantVillage	99.35%	No treatment, no user management
Sladojevic et al. (2016)	Deep learning (Caffe)	96.3%	Only web demo, no history
Ferentinos (2018)	Multiple CNNs	99.53%	Model-centric, no application

Gaps identified:

- No end-to-end system (user management, treatment, history)
- No admin dashboard, no OTP password reset

Our project fills these gaps.

FUNCTIONAL REQUIREMENTS

- **User Management:** Register, login, JWT, change password, OTP forgot password
- **Disease Detection:** Upload image, preprocess, predict disease + confidence + treatment
- **Scan History:** Chronological list of past predictions
- **Admin:** Dashboard stats, view all users, activate/deactivate, change role, logs

NON-FUNCTIONAL REQUIREMENTS

Requirement	Target
Prediction response time	≤3 seconds
Model accuracy	≥85% (target 90-95%)
JWT token expiry	24 hours
File upload limit	10 MB (jpg, jpeg, png)
ML memory usage	≤2 GB RAM
Error logging	Console + user-friendly messages

SLIDE 9: METHODOLOGY (AGILE – 3 SPRINTS)

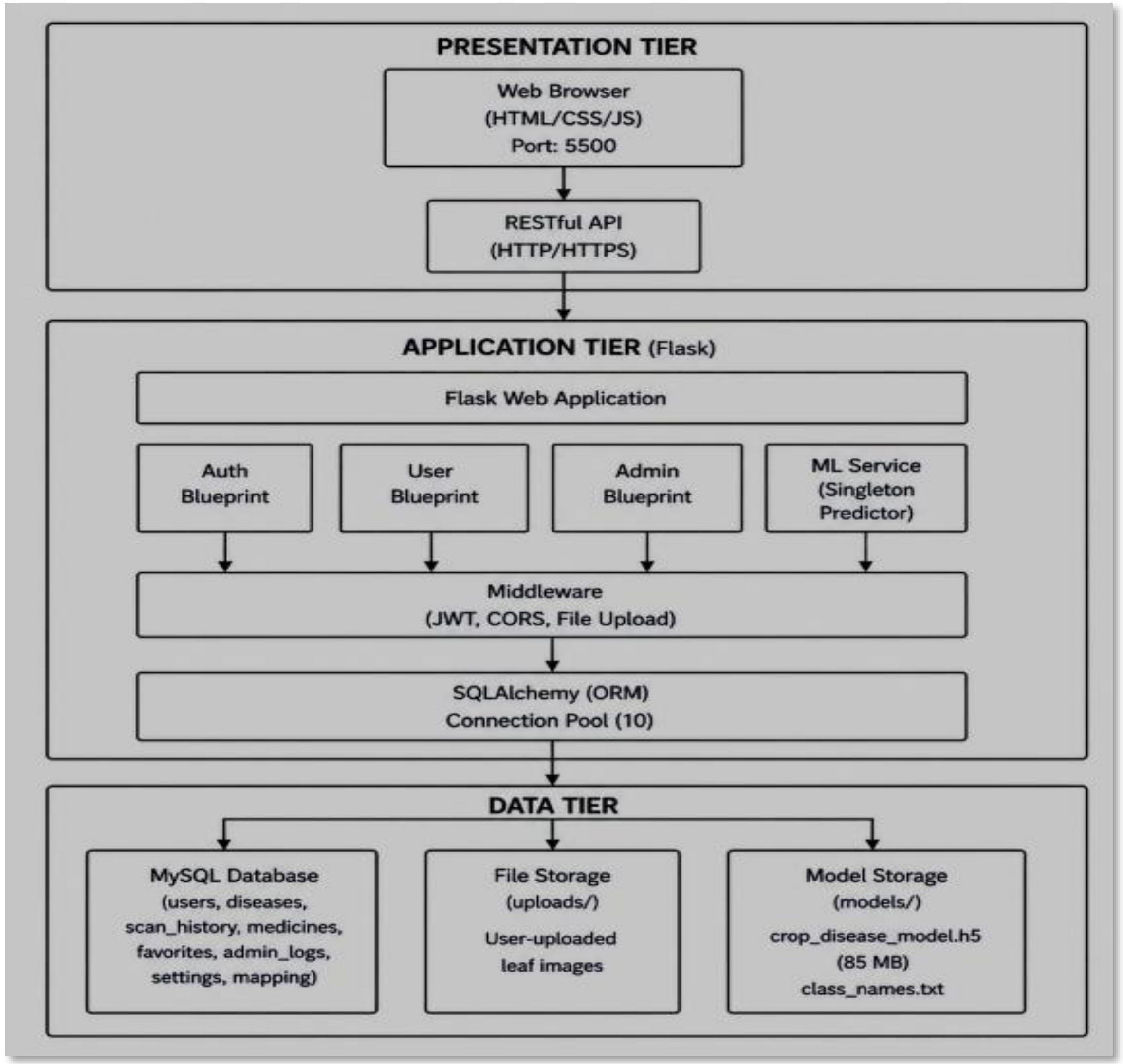
Sprint	Duration	Activities
1	Weeks 1-4	Requirements, DB design, Flask auth APIs, basic web pages
2	Weeks 5-8	Disease library (38), prediction (mock), web dashboard, admin dashboard
3	Weeks 9-12	Model training (MobileNetV2), real prediction, testing, documentation

Tools: Git, VS Code, Postman, MySQL Workbench, TensorFlow, Kaggle

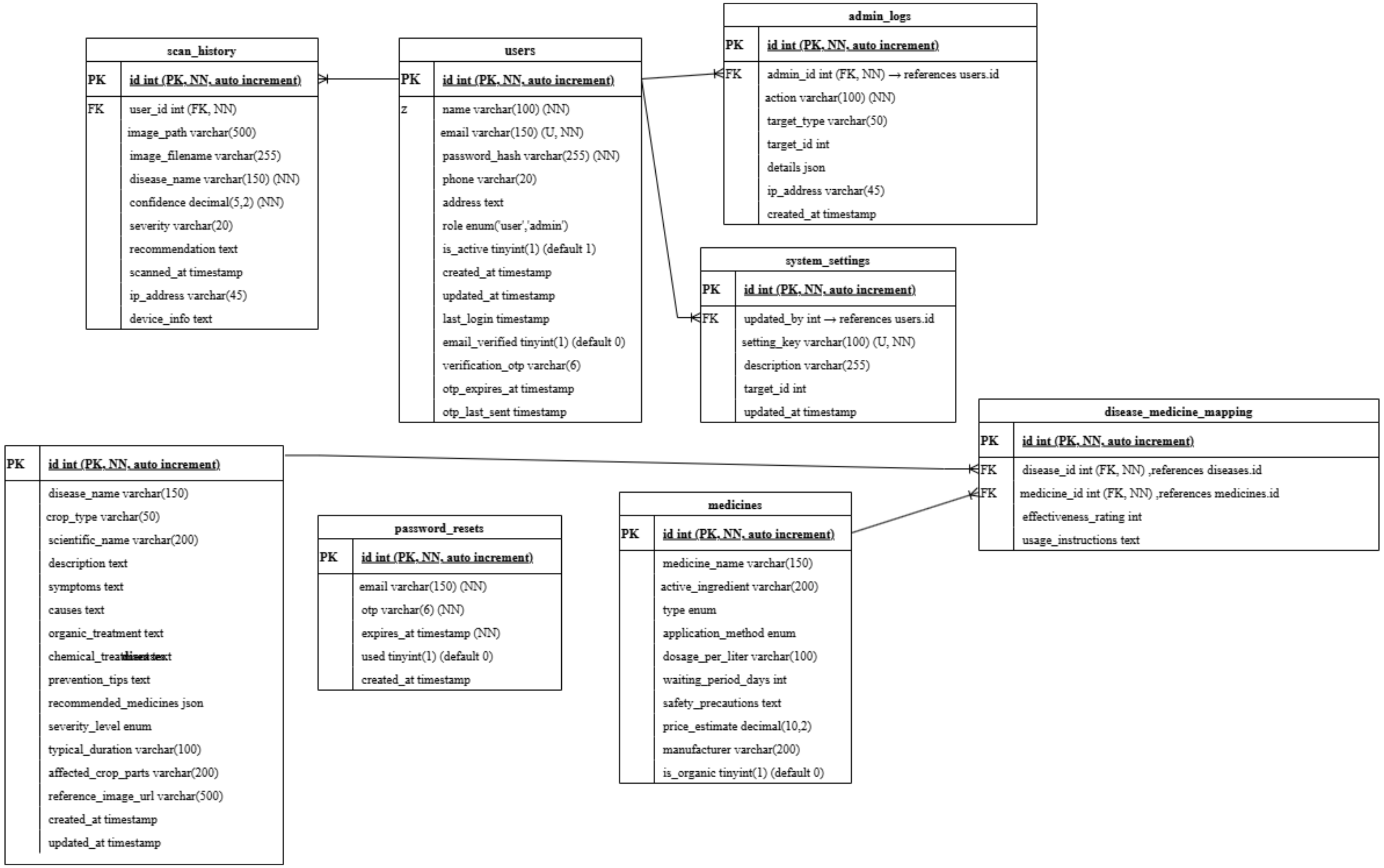
ML PIPELINE

- **Dataset:** PlantVillage (54,303 images, 38 classes, 14 crops)
- **Preprocessing:** Resize 224×224, normalise, augment (rotation, shifts, flips)
- **Model:** MobileNetV2 (pre-trained) + custom head (global pooling, dropout, dense 512, dense 38)
- **Training:** Two-phase transfer learning
    - Phase 1: 10 epochs, LR 0.001 (base frozen)
    - Phase 2: 10 epochs, LR 0.00001 (last 30 layers unfrozen)
- **Target:** Accuracy ≥90%, inference <2 seconds

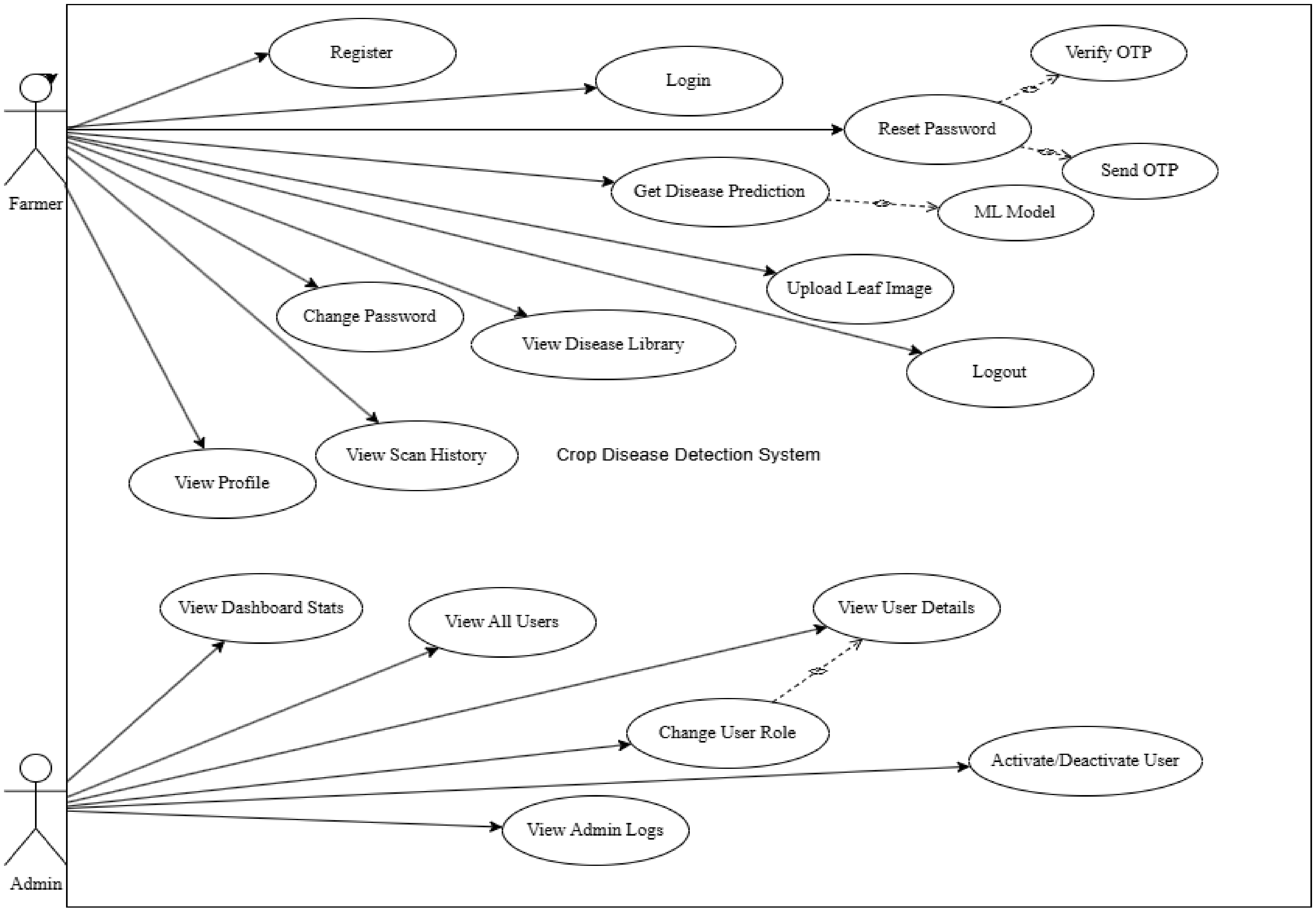
SYSTEM ARCHITECTURE (3-TIER)



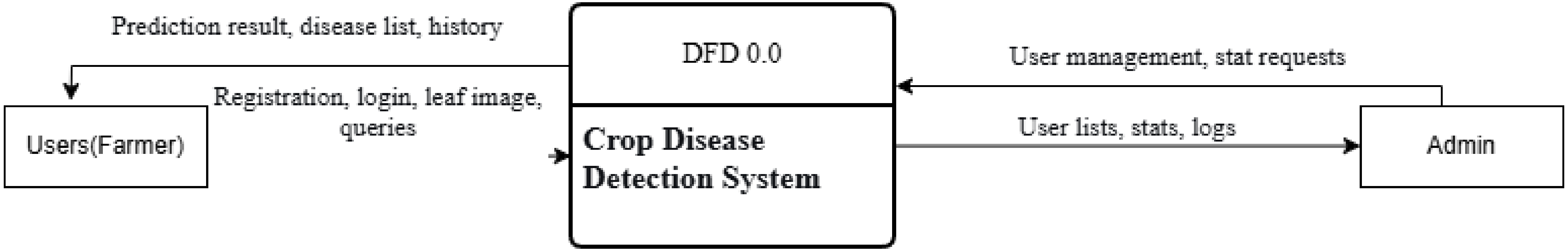
ER DIAGRAM (KEY TABLES)



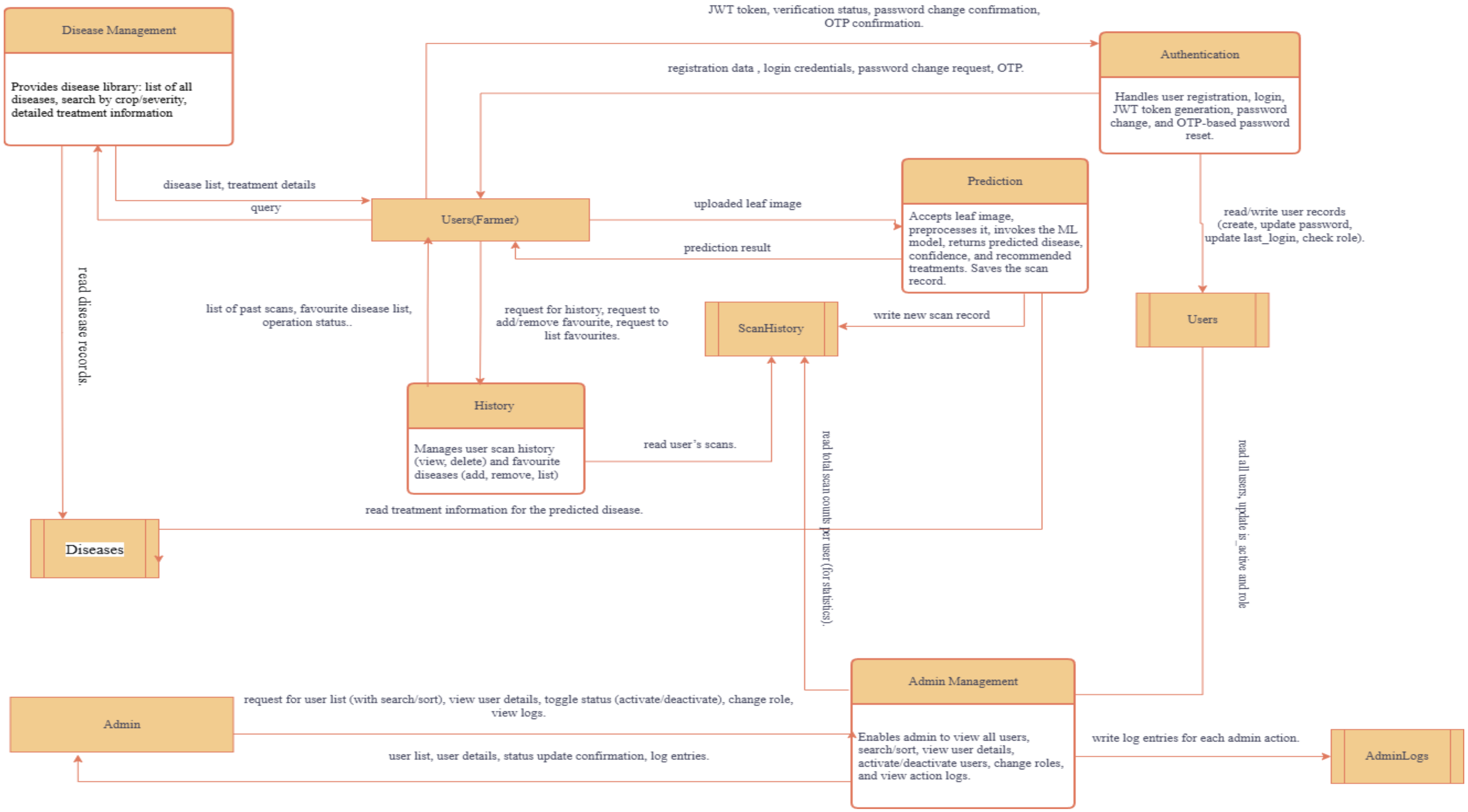
USE CASE DIAGRAM



DATA FLOW DIAGRAMS (DFD) (0)



DATA FLOW DIAGRAMS (DFD)



COST ESTIMATION (COCOMO II)

Component	Amount (NPR)
Salaries (4 persons × 3 months)	3,30,000
Laptop depreciation	28,750
Infrastructure & overheads	12,000
Contingency (10%)	35,900
Total	**4,06,650 (≈4.07 lakhs)**

- Based on 7.2 KSLOC → 19 person-months effort
- Team: 4 persons, 3 months
- Estimate may vary; for academic planning only

PROPOSED DELIVERABLES

- **Complete System** – Web application + REST API
- **Project Report** – Full documentation (this document)
- **Source Code Package** – Backend, frontend, database scripts, trained model
- **Presentation Slides** – This PPT,Report PPT

CONCLUSION

- Successfully integrates MobileNetV2 with web + API
- Provides rapid, accessible disease diagnosis and treatment
- All functional & non-functional requirements met
- Cost estimated ₹4.07 lakhs for 4 persons / 3 months
- Ready for deployment; future work: offline support, more crops, multi-language

THANK YOU